Problem sheet 11 for Rigid Analytic Geometry Summer Term 2026

For the complete description of van der Put point of $X = \mathrm{Sp}\mathbb{T}_1$, it is still necessary to deal with $\xi \in X^*$ for which $R = R_\xi \in |K^*|$, $B = B_{\leq R_\xi}(\xi)$ exists, and $\mathcal{S}(B) \not\subseteq \xi$. We have seen in Problem 3 from the previous sheet that $\mathcal{S}(B)\setminus\xi$ has a unique element S in this case. By translation by an element of $B \setminus S \subseteq K^\circ$ we may assume $B = B_{\leq R}(0)$ and $0 \notin S$. Thus, $S = \big\{x \in K \;\big|\; |x| = R\big\}$.

The following valuation of $\mathbb{T}_1$ turns out to be associated with this situation. For $f = \sum_{i=0}^\infty f_i T^i \in \mathbb{T}_1 \setminus 0$, let

$$n = \max_{i\in\mathbb{N}} R^i\,|f_i| \tag{1}$$

and

$$j = \min\big\{i \in \mathbb{N} \;\big|\; R^i\,|f_i| = n\big\}.$$

For $f \in \mathbb{T}_1$, let $N(f) = (n, j)$ with n and j from 1 if $f \neq 0$, and $N(0) = 0$. This is a continuous valuation

$$\mathbb{T}_1 \xrightarrow{N} \big((0,\infty)_\mathbb{R} \times \mathbb{Z}\big) \cup \{0\}$$

with the target group ordered by

$$(n, j) \preceq (\nu, \iota) \text{ if and only if } n < \nu \text{ or } n = \nu \text{ and } j \geq \iota.$$

This is similar to the ordering described after sheet 9.(1–2) in the sense that the first factor is most significant in both cases, but the ordering of the second one is reversed compared with sheet 9.

Problem 1 (3 points)**.** *For $f \in \mathbb{T}_1 \setminus \{0\}$ and (n, j) as in (1), there is $r \in (0, R)_\mathbb{R}$ such that*

$$|f(x)| = n\Big(\frac{|x|}{R}\Big)^j$$

when $r < |x| < R$.

Problem 2 (5 points)**.** *Consider the valuation N from the previous problem, and let $(f_i)_{i=0}^m \in \mathbb{T}_1^{m+1}$ such that $\mathbb{T}_1 = \langle f_0, \ldots, f_m\rangle_{\mathbb{T}_1}$, and let $\Omega = \mathcal{R}_X(f_0|f_1, \ldots, f_m) \subseteq K^\circ$. Show the following:*

- *If $N(f_i) \leq N(f_0)$ for $1 \leq i \leq m$, then there is $r \in (0, R)_\mathbb{R}$ such that $B_{\leq R}(0) \subseteq \Omega \cup B_{\leq r}(0) \cup S$.*
- *If there is $i \in [1, m]_\mathbb{N}$ such that $N(f_0) < N(f_i)$, then there is $r \in (0, R)_\mathbb{R}$ such that $\Omega \cap B_{\leq R}(0) \subseteq B_{\leq r}(0) \cup S$.*

Moreover, show that there is a unique van der Put point $\xi \in X^*$ such that $\xi \cap \mathfrak{Rat}_X$ is the set of Ω for which the first point in the above alternative occurs.

Problem 3 (2 points)**.** *Let $\xi \in X^*$ such that $R := R_\xi$ is an element of $(0,1]_{\mathbb{R}} \cap |K^\times|$, such that $B_{\leq R}(0) \in \xi$ while $S = \left\{x \in K \;\middle|\; |x| = R\right\} \notin \xi$. Show that ξ coincides with the van der Put point from the previous problem!*

Let K be any field equipped with an ultrametric absolute value $|\cdot|$. As on sheet 2, by a *normed K-vector space* we understand a K-vector space V with a map $V \xrightarrow{\|\cdot\,|V\|} [0,\infty)_{\mathbb{R}}$ such that the following holds for $x, y \in V$ and $\lambda \in K$:

- $\|x+y\,|V\| \leq \max(\|x\,|V\|\,, \|y\,|V\|)$
- $\|\lambda x\,|V\| = |\lambda|_K \, \|x\,|V\|$
- $\|x\,|V\| = 0$ implies $x = 0$.

A linear functional $V \xrightarrow{\ell} K$ is called *bounded* if there is $B \in [0,\infty)_{\mathbb{R}}$ such that $|\ell(v)| \leq B\,\|v\,|V\|$. We define V^* as the K-vector space of such functionals and $\|\ell\,|V^*\|$ as the infimum of the set of such B.

Definition 1. *We say that the* Hahn-Banach theorem *holds for K if for every subspace W of a normed K-vector space V, every $\ell \in W^*$ has an extension to a bounded linear functional $L \in V^*$ such that $\|L\,|V^*\| = \|\ell\,|W^*\|$.*

Problem 4 (2 points)**.** *Show that the Hahn-Banach theorem holds for K if and only if it holds in the case where W is of (algebraic) codimension 1 in V.*

Definition 2. *A* Banach limit *for K is a K-linear functional ɬ on the space ℓ_∞ of sequences $x = (x_i)_{i=1}^\infty$ of elements of K for which*

$$\|x\,|\ell_\infty\| = \sup_{1 \leq i < \infty} |x_i|_K$$

is finite such that $\|\text{ɬ}\,|\ell_\infty^*\| = 1$ and such that $\ell(x) = \xi$ if $x \in \ell_\infty$ and $\xi \in K$ are such that $x_i = \xi$ for all sufficiently large natural numbers ξ.

Problem 5 (1 point)**.** *If the Bahn-Banach theorem holds for K then Banach limits exist for K.*

Problem 6 (2 points)**.** *If Banach limits exist for K then K is spherically complete.*

In the case where $K = \mathbb{R}$, the classical (Archimedean) Hahn-Banach theorem is proved in the following way, after reducing to the case where $\operatorname{codim}(W, V) = 1$ in a similar way as in 4. Let $v_o \in V \setminus W$. The extension L to V of a linear functional ℓ on W is uniquely determined by $\lambda = L(v_o)$. For each $w \in W$, the condition $|\ell(w + v_o)| \leq \|\ell\,|W^*\| \cdot \|w + v_o\,|V\|$ leads to an upper bound B_w and a lower bound b_w for λ,

and one checks that $b_w \leq B_{\tilde{w}}$ holds for every pair of elements of W. By the compactness of bounded closed intervals in $\mathbb{R}$, it follows that these infinitely many constraints on λ are compatible.

This strategy works in the non-Archimedean case as well but not without further assumptions on K. Obviously K must be complete as the Banach limit of a Cauchy sequence is its limit. But the previous problem shows that spherical completeness is also needed.

Problem 7 (5 points)**.** *Assume that K is spherically complete in the sense explained on sheet 9. Show that the Hahn-Banach theorem holds for K!*

The result of these considerations is

Theorem 1. *For a field K equipped with a non-Archimedean absolute value, the following conditions are equivalent:*

- *K is spherically complete.*
- *The Hahn-Banach theorem holds for K.*
- *K has Banach limits.*

For instance, the Hahn-Banach theorem holds for $\mathbb{Q}_p$, or any of the local fields considered in basic algebraic number theory. But it does not hold for $\mathbb{C}_p$.

Solutions should be e-mailed to your tutor s45qhou at uni hyphen bonn dot de before Monday July 13. Good luck for this last exercise sheet, and good luck for the exam!